Ibn Tofail University

Probability and Statistics

Problem Set II

Exercise 1:

Consider a cubic die whose faces are numbered from 1 to 6, biased such that the probability of obtaining face number k is proportional to k, with proportionality coefficient p. We roll the die once and denote by X the number on the face obtained.

1. Determine the probability distribution of X (as a function of p).
2. Determine the value of p.
3. Calculate $E(X)$.
4. Let $Y = \frac{1}{X}$. Determine the distribution of Y and $E(Y)$.

Correction

Exercise 2:

A straight track is divided into squares numbered 0, 1, 2, ..., n, ... from left to right. A flea moves to the right, randomly jumping 1 or 2 squares at each move. Initially, it is on square 0. Let X_n be the random variable equal to the number of the square occupied by the flea after n jumps.

1. Determine the probability distribution of X_1, and calculate $E(X_1)$ and $V(X_1)$.
2. Let Y_n be the random variable equal to the number of times the flea jumped one square during the first n jumps. Determine the distribution of Y_n, then $E(Y_n)$ and $V(Y_n)$.
3. Express X_n as a function of Y_n and deduce the probability distribution of X_n, then $E(X_n)$ and $V(X_n)$.

Correction

Exercise 3:

We play heads or tails with a biased coin. In each toss, the probability of getting tails is $\frac{1}{3}$. The tosses are assumed to be independent. Let X be the random variable equal to the number of tosses needed to obtain two consecutive heads for the first time. Let n be a non-zero natural integer. We denote p_n as the probability of the event $[X = n]$. Additionally, we denote F_i as the event "obtaining tails on the i-th toss".

1. Express the events $[X = 2]$, $[X = 3]$, $[X = 4]$, $[X = 5]$ in terms of the events F_i and $\overline{F_i}$. Determine the values of p_1, p_2, p_3, p_4, p_5.
2. Using the total probability formula, distinguishing two cases based on the result of the first toss, show that:

$$\forall n \geq 3, \quad p_n = \frac{2}{9}p_{n-2} + \frac{1}{3}p_{n-1}$$

3. Deduce the expression of p_n as a function of n, for $n \geq 1$.
4. Calculate $E(X)$.

Correction

Exercise 4:

An urn contains 2 white balls and $n-2$ red balls. We perform n draws without replacement from this urn. Let X be the rank (draw number) of the first white ball drawn and Z be the rank of the second white ball drawn.

1. Determine the joint distribution of the pair (X, Z).
2. Deduce the distribution of Z.

Correction

Exercise 5:

We have n boxes numbered from 1 to n. Box k contains k balls numbered from 1 to k. We randomly choose a box and then a ball from that box. Let X be the box number and Y be the ball number.

1. Determine the joint distribution of (X, Y).
2. Calculate $P(X = Y)$.
3. Determine the distribution of Y and $E(Y)$.

Correction

Exercise 6:

Let $(X_i)_{i\in\mathbb{N}}$ be a sequence of independent Bernoulli random variables with parameter p. Let $Y_i = X_i X_{i+1}$.

1. What is the distribution of Y_i?
2. Let $S_n = \sum_{i=1}^{n} Y_i$. Calculate $E(S_n)$ and $V(S_n)$.

Correction

Exercise 7:

Let (X, Y) be a pair of discrete random variables with values in $\mathbb{N}^2$, with joint distribution:

$$P([X = i] \cap [Y = j]) = \frac{a}{2^{i+1} j!}$$

1. Determine all possible values of a.
2. Determine the marginal distributions.
3. Are X and Y independent?

Correction

Exercise 8:

Let $X_1, \ldots, X_n$ be finite discrete random variables. Show that:

$$E(X_1 + \cdots + X_n) = \sum_{i=1}^{n} E(X_i).$$

Correction